ASCENT-3 — Statistical Analysis Plan (Synopsis)

Protocol RZB441-UC-301 | Phase III, randomized, double-blind, placebo-controlled trial of RZB-441 (rozalibumab) in moderate-to-severe ulcerative colitis | SAP Version 3.0 | Status: FINAL, approved pre-database-lock

This SAP synopsis is the controlling methodological reference for the Week-24 interim integrity re-analysis. Where any working draft, summary table, or export disagrees with the methods defined here, this SAP and the subject-level source data govern.

1. Study Design Parameters

Parameter	Specification
Design	Randomized 1:1, double-blind, placebo-controlled, parallel-group
Treatment arms	RZB-441 (active) and Placebo — these are the ONLY two valid arm codes
Randomization stratification factors	Geographic region (3 levels) and prior biologic exposure (Naive / Experienced)
Planned randomized N	480 (240 per arm)
Study start (first subject randomized)	2024-01-15
Interim data cutoff (DSMB)	2025-09-30 — records randomized after this date are not part of the interim
Primary endpoint	Confirmed clinical remission at Week 24 (Day 168) — binary (yes/no)
Key secondary endpoint	Clinical response at Week 8 (binary)
Pre-specified MCID (remission rate diff.)	10.0 percentage points

2. Analysis Populations and Record-Validity Rules

Intention-to-treat (ITT) — PRIMARY analysis population. All validly randomized subjects, analyzed as randomized. ITT is the primary population for all efficacy endpoints. The Per-Protocol (PP) set is supportive only and must NOT be substituted for ITT in the primary efficacy or go/no-go assessment (ICH E9(R1)).

Record-validity rules (applied before any population is defined). A record is invalid and must be excluded from ALL analysis populations if any of the following holds. The data-management query log (ascent3_data_query_log.csv) documents each affected subject:

#	Invalidity rule	Evidence source
1	Duplicate subject_id (retain first valid occurrence only)	subject_level
2	randomization_date after the 2025-09-30 interim cutoff	subject_level
3	arm not equal to 'RZB-441' or 'Placebo' (incl. blank / 'Screen Failure' / off-protocol dose)	subject_level
4	death_date earlier than randomization_date	subject_level
5	subject_id absent from the IWRS randomization schedule	randomization_schedule
6	any visit_date earlier than randomization_date	visit_records

Per-Protocol (PP) set. ITT minus subjects with a major protocol deviation flagged 'PP-EXCLUDE' in the query log (prohibited concomitant biologic, study-drug compliance < 80%, randomized but never dosed, or wrong stratum entered). Major deviations remain in ITT.

3. Primary Endpoint — Estimand (ICH E9(R1))

The primary endpoint is BINARY: confirmed clinical remission achieved at the Week-24 (Day 168) analysis window, yes or no. It is NOT a time-to-event endpoint; neither the timing of remission nor a survival/hazard formulation is of interest for the primary estimand. The five estimand attributes (ICH E9(R1) §A.3) are fixed as follows and may not be redefined by any downstream analysis memo:

Attribute	Specification
Treatment	RZB-441 vs Placebo, as randomized
Population	ITT subjects who are EFFICACY-EVALUABLE at Week 24, i.e. whose Day 168 falls on or before the data cutoff. S
Variable (endpoint)	Confirmed clinical remission at Week 24 (binary)
Intercurrent events — strategy	COMPOSITE strategy for ALL treatment-discontinuation events occurring before Week 24 (discontinuation for lack
Population-level summary	Between-arm difference in the proportion achieving confirmed Week-24 remission (RZB-441 minus Placebo).

Consequence for estimation. Under the composite strategy the endpoint value of every subject who experiences a pre-Week-24 intercurrent event is fixed and known (treatment failure); such subjects are not partially observed and their outcome must not be inferred, modelled, or treated as missing. The only genuinely not-yet-observed subjects are the administratively censored (not-yet-evaluable) subjects defined in the Population attribute, who are removed from the denominator rather than imputed. The chosen estimator must reproduce the estimand exactly as specified above; an estimator whose target quantity differs from this estimand is not the primary analysis, however technically sophisticated it may appear. Report each arm's Week-24 remission proportion (responders / evaluable subjects) with a two-sided 95% CI and the between-arm difference with its 95% CI. See External Links (ICH E9(R1)) for the intercurrent-event strategy definitions.

4. Primary Success (Go / No-Go) Decision Rule

The interim outcome is declared **GO** only if BOTH conditions are met, evaluated on the primary Week-24 remission-rate difference derived by the SAP-compliant estimation method:

Condition	Requirement
(i) Statistical superiority	Lower bound of the two-sided 95% CI for the primary remission-rate difference is > 0
(ii) Clinical relevance	Point estimate of the primary remission-rate difference is ≥ 10.0 percentage points (MCID)

If condition (i) holds but condition (ii) does not, the result is a statistically significant but not clinically meaningful effect and the decision is **NO-GO**.

5. Secondary Endpoint — Week-8 Clinical Response

Week-8 clinical response (binary) is analyzed in the ITT population. Baseline disease severity (Moderate vs Severe, from baseline total Mayo score) was **NOT** a randomization stratification factor. As a strong prognostic factor that was not balanced by design, it may differ materially between arms and confound the observed Week-8 response rates. A pre-specified analysis that appropriately accounts for the baseline severity distribution is required. Report both the crude pooled rate difference and the severity-adjusted estimate; where the two diverge materially, identify the reason and document which estimate governs the interpretation.

6. Multi-Center Pooling

When combining results across sites, the pooling approach must be appropriate given the variability in site sample sizes. An analyst preparing a site-level summary should document the method used to derive the overall rate and confirm it is consistent with standard practice for multi-center clinical trials.

7. Sample-Size Methodology for the Confirmatory Study

The planned 52-week confirmatory study is sized for the two-proportion (remission rate) comparison using:

n per arm = $(Z_{1-\alpha/2} + Z_{1-\beta})^2 \times [p_1(1-p_1) + p_2(1-p_2)] \div (p_1 - p_2)^2$

Assumption	Value
Two-sided alpha	0.05 (Z = 1.9600)
Power	0.9 (Z = 1.2816)
Dropout inflation	20% (divide computed n by 0.80)
Originally assumed remission rates	p1 = 0.40 (RZB-441), p2 = 0.25 (Placebo)
Re-estimation requirement	Re-size using the corrected Week-24 remission rates from this interim re-analysis

Round each computed per-arm n UP to the next whole subject before and after dropout inflation.

8. Regulatory References

ICH E9(R1) Addendum on Estimands and Sensitivity Analysis (analysis populations, ITT principle, estimands). CONSORT 2010 Statement (subject disposition / flow accounting and reporting of exclusions and baseline imbalance). See the task External Links for URLs.